

Group A		
	CW	HW
1. Definition with classifications		
2. Formation		
3. Order and Degree		
4. Solution with method		
4.1) Variable Separable PDF page : 4	$\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$	Book Page 6 –(ex 1,3,4,5)
4.1.1) Reducible to variable separable PDF page : 7	$\frac{dy}{dx} = (4x + y + 1)^2$	Book Page 8 –(ex 2,3,4,5,7)
4.2) Homogenous DEs PDF page : 8	$(x \cos \frac{y}{x} + y \sin \frac{y}{x})y = (y \sin \frac{y}{x} - x \cos \frac{y}{x})x \frac{dy}{dx}$	Book Page 10 –(ex 1,2,3,4)
4.2.1) Reducible to HDEs		
4.2.2) Special case to HDEs PDF page : 14	$(2x - 2y + 5) \frac{dy}{dx} = x - y + 3$	Book Page 16 –(ex 5,6)
4.2.3) Particular case of HDEs PDF page : 15	$\frac{dy}{dx} = \frac{2x-3y+5}{3x-5y+2}$	Book Page 17 –(ex 1,2,3,4)
4.3) Linear DEs PDF page : 19	$(1 - x^2) \frac{dy}{dx} - xy = 1$	Book Page 19 –(ex 2(a),(b),3), Page 21 –ex 8, Page 22–(ex 2,3,4)
4.3.1) Linear DE when X is dependent		
4.3.2) Reducible to linear Des/Bournoli's DEs PDF page : 23	$\frac{dy}{dx} = x^3 y^3 - xy$	Book Page 24 –(ex 2,3,4,6)
4.4) Exact DEs PDF page : 26	$(y^4 + 4x^3y + 3x)dx + (x^4 + 4xy^3 + y + 1)dy = 0$	Book Page 37 –(ex 1), Page 38 – (ex 2,3) Page 39 – ex 6)
4.4.1) Reducible to DEs		
4.4.2) Finding IF rule PDF page : 30	$(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$	Rules 1 – Page 45(4,5) Rules 2 – Page 46(8,9) Rules 3 – Page 47(1,2) Rules 4 – Page 49(4)
5. Linear differential equation with constant co-efficient (CF,PI) Definition from book page -60 Theorem & auxiliary equations from pdf PDF page : 33	Case I : $\left(\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y\right) = 0$ Case II : $(D^2 - 2D + 1)y = 0$ Case III : $(y_2 + y_1 + y) = 0$	Book Page 64 –(ex 1,2,3,4) Page 75- (ex 1,2,3,4,5,6 (a),(c),7,9)
5.1) Complementary Function		
5.2) Particular Integral	$\left(\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 2y\right) = e^x + \cos x$	

Group B		
	CW	HW
1)Bessel Function/Equations PDF page : 42	Q1. Define Bessel's Equation Q2. Define Bessel's Function Q3. Using Bessel's Function Prove That : $a) J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ $b) J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$	
2)Legendre's Function/Equations PDF page : 61	Q1. Define Legendre's Equation Q2. Define Legendre's Function Q3. Show That : $a) P_n(1) = 1$ $b) P_n(-1) = (-1)^n$ Q4. Define Rodrigue's Formula. Using this evaluate, $a) P_0(x)$ $b) P_1(x)$ $c) P_2(x)$ $d) P_3(x)$ $e) P_4(x)$ Q5. Prove that, $\int_{-1}^1 [P_n(x)] dx = \begin{cases} 0, & n \neq 0 \\ 2, & n = 0 \end{cases}$	
3)Method of variation of parameters Theorem from PDF PDF page : 78	$y'' + y = \operatorname{cosec} x$	PDF page no : 81 3(a),1(a),2
4)Method of undetermined co-efficient		
5)Linear Partial Differential EQn of order one : Definition from pdf - PDF page : 97+102		
6)Lagrange's Method Definition from pdf PDF page : 102	$xz_p + yz_q = z^2$	Book Page – 07 Ex 2,3,4,7(a),(b),9
6)Char pits method		
7)Application of DEs in real word problem	3.17,3.18,3.20,3.26	